
Additional Problems for Ch 16 HRW – Waves 1
 Addisionele Probleme vir Hfst 16 HRW – Golwe 1

	9 th Edition	8 th Edition
Problems	9, 12, 32, 40, 54, 55	11, 10, 32, 44, 56, 57

9	11	(a) $y_m = 3.0$ mm. (b) $k = 16$ rad/m . (c) $\omega = 3.2 \times 10^2$ rad/s. (d) We choose the minus sign (between kx and ωt) in the argument of the sine function because the wave is shown traveling to the right [in the $+x$ direction] – see section 16-5). Therefore, with SI units understood, we obtain $y = y_m \sin(kx - \omega t) \approx 0.0030 \sin(16x - 3.2 \times 10^2 t)$.	(a) $y_m = 3.0$ mm. (b) $k = 16$ rad/m . (c) $\omega = 2.4 \times 10^2$ rad/s. (d) minus sign
12	10	4.24 m/s	
32	32	(a) $\phi = 2 \cos^{-1} \left(\frac{1.50 y_m}{2 y_m} \right) = 82.8^\circ$. (b) Converting to radians, we have $\phi = 1.45$ rad. (c) In terms of wavelength (the length of each cycle, where each cycle corresponds to 2π rad), this is equivalent to $1.45 \text{ rad} / 2\pi = 0.230$ wavelength.	
40	44	$\lambda = 10$ cm	
54	56	(a) $y_m = \frac{1}{2}(9.0 \text{ mm}) = 4.5$ mm, due to the fact that the amplitude of the standing wave is $\frac{1}{2}(1.80 \text{ cm}) = 0.90 \text{ cm} = 9.0$ mm. (b) $k = 2\pi/\lambda \approx 16 \text{ m}^{-1}$. (c) $T = 12$ ms therefore $\omega = 5.2 \times 10^2$ rad/s. (d) The two waves are therefore $y_1(x, t) = (4.5 \text{ mm}) \sin[(16 \text{ m}^{-1})x + (520 \text{ s}^{-1})t]$ and $y_2(x, t) = (4.5 \text{ mm}) \sin[(16 \text{ m}^{-1})x - (520 \text{ s}^{-1})t] .$ If one wave has the form $y(x, t) = y_m \sin(kx + \omega t)$ as in y_1, then the other wave must be of the form $y'(x, t) = y_m \sin(kx - \omega t)$ as in y_2. Therefore, the sign in front of ω is minus.	
55	57	$\Delta x = (1.00 \text{ m/s})(0.250 \text{ s}) = 0.250$ m.	

Additional Problems for Ch 17 HRW – Waves II
 Addisionele Probleme vir Hfst 17 HRW – Golwe II

	9 th Edition	8 th Edition
Problems	16, 45, 48, 51, 55, 61, 63	16, 41, 46, 51, 57, 63, 59

16	16	$\Delta\phi = 4.12 \text{ rad.}$	
45	41	(a) pipe <i>A</i> is open at both ends (b) pipe <i>B</i> is open at only one end	
48	46	(a) 260 Hz. (b) Since $f_n = nf_1$ then $n = 260/65 = 4$. (c) $f = 840 \text{ Hz.}$ (d) Since $f_n = nf_1$ (for n odd) then $n = 840/120 = 7$.	
51	51	$T = 1.98 \times 10^{-3} \text{ s.}$	$T = 2.25 \times 10^{-3} \text{ s.}$
55	57	(a) = 521 Hz (b) = 560 Hz	(a) = 526 Hz (b) = 555 Hz
61	63	$f = 4.06 \times 10^4 \text{ Hz}$	$f = 4.1 \times 10^4 \text{ Hz}$
63	59	$f = 200 \text{ Hz}$	$f = 155 \text{ Hz}$